

BS Computer Science Suggested Four Year Plan
2023 – 2024 Catalog

Freshman

Description	Course No
Introduction to CS&SE	CS 1200
Introduction to Engr'g. & CS	ECS 1100
Calculus I	Math 2413 or 2417
Programming Fundamentals	CS 1436
6 hours CORE: Choices below	
Freshman Seminar	UNIV 1010

Diagram showing Freshman course flow: ECS 1100 leads to UNIV 1010 and PHYS 2325. *MATH 2413 or *MATH 2417 leads to MATH 2414 or MATH 2419. CS 1200 leads to **CS 1436. **CS 1436 leads to **CS 1337. CORE courses are also listed.

Freshman

Description	Course No
Computer Science I	CS 1337
Discrete Mathematics I	CS 2305
Calculus II	Math 2414 or 2419
Mechanics and Heat	Phys 2325
Mechanics and Heat Lab	Phys 2125
Computer Science II	CS 2336 or CS 2337
3 hours CORE: Choices below	

Diagram showing Freshman course flow: PHYS 2325 leads to PHYS 2326. MATH 2414 or MATH 2419 leads to **CS 2305. **CS 2305 leads to **CS 3305 and CS 2340. **CS 1337 leads to **CS 2336 or **CS 2337. CORE courses are also listed.

Sophomore

Description	Course No
Discrete Math for Computing II	CS 3305
Computer Architecture	CS 2340
Computer Science II	CS 2336 or 2337
Electricity and Magnetism	Phys 2326
Electricity and Magnetism Lab	Phys 2126
3 hours CORE: Choices below	

Diagram showing Sophomore course flow: PHYS 2326 leads to PHYS 2126. **CS 3305 leads to CS 3341 and CS 3345. CS 2340 leads to CS 3345. **CS 2336 or **CS 2337 leads to CS 3377. CORE courses are also listed.

Sophomore

Description	Course No
Probability & Statistics in CS	CS 3341
Algorithm Analysis & Data Struc	CS 3345
Systems Programming in UNIX and Other	CS 3377
Linear Algebra	Math 2418
3 hours CORE: Choices below	

Diagram showing Sophomore course flow: Math 2418 leads to CS 3341. CS 3341 leads to CS 3345. CS 3345 leads to CS 3377. CORE courses are also listed.

Junior

Description	Course No
Software Engineering	CS 3354
Programming Language Paradigms	CS 4337
Digital Logic and Computer Design	CS 4341
Digital Systems Laboratory	CS 4141
Professional & Technical Comm	ECS 3390
3 hours CORE: Choices below	
Core Curriculum Assessment	UNIV 2020

Diagram showing Junior course flow: CS 4341 and CS 4141 lead to CS 4337. CS 3354 leads to CS 3345. ECS 3390 leads to CS 3345. CORE courses are also listed.

Junior

Description	Course No
Prof. Responsibility in CS and SE	CS 3162
Operating Systems Concepts	CS 4348
Adv Data Structures & Algorithms	CS 4349
3 hours CORE: Choices below	
3 hours Free Elective	See Advisor

Diagram showing Junior course flow: CS 3345 leads to CS 4349 and CS 3162. CS 4348 leads to CS 4349. CORE courses are also listed.

Senior

Description	Course No
Automata Theory	CS 4384
CS Guided Elective	CS 43XX
CS Guided Elective	CS 43XX
Database Systems	CS 4347
1 hours Free Elective	See Advisor

Diagram showing Senior course flow: CS 4384 leads to CS 4347. CS 4347 leads to CS 4485. CS 4485 leads to CS 43XX. CORE courses are also listed.

Senior

Description	Course No
Computer Science Project	CS 4485
CS Guided Elective	CS 43XX
6 hours Free Elective	See Advisor

Diagram showing Senior course flow: CS 4485 leads to CS 43XX. CORE courses are also listed.

*A grade of C- or better is needed in Math 2413 or Math 2417 to move forward.
**A grade of C or better is required for CS 1336, CS 1337, CS 2305, CS 2336, CS 2337 and CS 3305.

Table with 5 columns: Rhets, 3 hours From, 3 hours From, 6 hours From, 3 hours From. It lists various core choices and their prerequisites.

Courses may be taken in any order, as long as prerequisites are met.

LEGEND
Prereq Critical Path
Pre-requisite
Pre- or Co-requisite